

MASTER'S COMPREHENSIVE EXAMINATION

Part I

Theory (90 minutes)

December 1, 2007

Instructions: Attempt any three out of the following four questions. Mention which three you would like to have graded. Open book and open notes.

1. Let $X \sim \text{Bin}(3, 0.5)$ and $Y \sim \text{Poisson}(1)$. Assume X and Y are independent.

(a) Find $P\{X \geq 1\}$.

(b) Find $P\{Y = 2|X = 2\}$.

(c) Find $P\{Y = 0|X + Y = 1\}$.

2. Let $f_X(x) = 1$ for $.5 \leq x \leq 1.5$, $f_Y(y) = \frac{1}{2}$ for $1 \leq y \leq 3$, and X, Y be independent.

(a) What is $E[X + Y]$?

(b) What is $\text{Var}[X + Y]$?

(c) What is $P\{X < Y\}$?

3. Let $X_1 \sim N(u_1, 3)$ and $X_2 \sim N(u_2, 4)$ with $\text{Cov}(X_1, X_2) = -1$. Let $\alpha = 0.05$ and test $H_0 : u_1 = u_2$ vs $H_a : u_1 \neq u_2$ by rejecting when $|X_1 - X_2| > C_\alpha$. Find C_α and the power of the test when $u_1 - u_2 = 4$.

4. Let 3, 1, 5 be a random sample from $\text{Poisson}(\lambda)$. Write down the likelihood function. Maximize it to obtain sample estimate $\hat{\lambda}$. Estimate Standard Error (SE) of $\hat{\lambda}$.